1. How the Model was Trained (The "Learning" Phase)

Think of training like teaching a child to recognize shapes.

* Step A (Extracting Dots): We took thousands of pictures from the internet (HaGRID). We didn't show the AI the whole
picture. Instead, we used MediaPipe to "draw 21 dots" on the hand in every picture.
* Step B (Numbers): We turned those dots into a list of numbers (coordinates). For example: "If the thumb is at (1,2)
and the pinky is at (5,5), this is an Open Hand."
* Step C (The Brain): We gave these numbers to a Neural Network (the "Brain"). We told it: "When you see these
numbers, the answer is 'Pointing'. When you see those numbers, the answer is 'OK'."
* Step D (The Result): After seeing 4,000 examples, the Brain became very smart. We saved that smart Brain into a
tiny file called keypoint_classifier.tflite.

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2. How it finds Gestures (The "Running" Phase)

When you run app.py or test_video.py, this is what happens 15 times every second:

* Step 1 (Vision): The camera takes one frame (a picture).
* Step 2 (Mapping): MediaPipe looks at that picture and instantly finds your hand. It connects the 21 dots (the skeleton you see on screen).
* Step 3 (Comparison): The app takes the coordinates of those 21 dots and sends them to your trained TFLite Brain.
* Step 4 (The Guess): The Brain looks at the dots and says: "I am 98% sure this is a Pointing gesture."
* Step 5 (Scenario Logic): Finally, the code looks at the "Global Scenario."
* If your hand is moving side-to-side, the code says: "Wait, those dots are moving! That's not just an open hand, that is Waving!"
* If you have two hands, it combines them to say: "Both hands are waving! That's Arms Waving!"
* **Step 6 (Output):You are very welcome! Here is a very simple explanation of how your project "thinks" and learns:

1. How the Model was Trained (The "Learning" Phase)

Think of this like teaching a child to recognize shapes:

1. Gather Photos: We took 4,000 photos of different hands from the internet (HaGRID dataset).
2. Convert to Dots: We didn't show the AI the actual colors or pixels. Instead, we used a tool (MediaPipe) to find 21
dots on every hand (like the knuckles and fingertips). We saved the (x, y) location of those dots as a list of numbers.
3. The Exam: we showed these lists of numbers to a Neural Network and told it: "When the dots look like this, it's a
Pointing finger. When they look like this, it's a Fist."
4. The Brain: The AI learned the patterns of those dots. Once it was smart enough, we saved its knowledge into a tiny
file called keypoint_classifier.tflite.

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2. How it Finds Gestures (The "Running" Phase)

This happens every time you run the app, 15 times every second:

1. See the Camera: The app looks at a single frame from your webcam or video.
2. Connect the Dots: MediaPipe instantly finds the 21 dots on your hand in the video.
3. Shrink & Center: The app "normalizes" the dots. This means it makes the hand the same size in the math, so it doesn't matter if your hand is very close to the camera or very far away.
4. Ask the AI: It sends those 21 dots to the trained brain (.tflite file) and asks: "What is this?"
5. The Guess: The AI looks at its saved patterns and says: "I am 95% sure that is a Pointing gesture."
6. The Rules: The app then checks your project's rules (for example: "If the hand is also moving up and down, change the name to 'Beckoning' instead").
7. Output: Finally, it prints the name on your screen!

In short: We turned hands into math (dots), taught the AI the patterns of that math, and now the AI uses those patterns to guess what you are doing in real-time!